

MORINGA SCHOOL

PLANT LEAF DISEASE CLASSIFIER PROJECT REPORT

By

LEAF IT TO US TEAM



Abstract

This project addresses the critical need for early plant disease detection using deep learning. The methodology involves using a dataset of 10 distinct disease classes from corn, potato, and tomato leaves. This data was prepared by removing 14 duplicate images and applying data augmentation to resolve a significant class imbalance. Two models—a custom-built Convolutional Neural Network (CNN) and a MobileNetV2 transfer learning model—were then trained and evaluated. The custom CNN model proved to be the superior architecture, achieving an outstanding

96% accuracy and a 0.96 Macro F1-score, outperforming the MobileNetV2. Given its high performance, the CNN model was selected for deployment in a Streamlit web application, providing an accessible, instant diagnostic tool for farmers.

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INTRODUCTION

Business Overview

Agriculture is essential for food security and economic growth worldwide, especially in developing countries where most people depend on farming for their livelihoods. However, crop diseases threaten agricultural productivity, leading to reduced yields, financial losses, and food insecurity. This project detects plant leaf diseases in corn, tomato,

and potato crops using image classification techniques. The goal is to develop a machine learning model that distinguishes between healthy and diseased leaves based on visual symptoms. Automating disease detection enables farmers to make timely decisions, control disease spread, and improve crop health, reducing reliance on manual inspection, which is slow, error-prone, and requires expert knowledge.

Problem Statement

Crop diseases such as **blight, leaf spots, and rust** significantly reduce the productivity of **corn, tomato, and potato** crops. Traditional detection methods rely on manual observation by farmers or agricultural experts. These methods are time-consuming, subjective, and prone to human error. A technological solution that accurately detects and classifies plant leaf diseases at an early stage is urgently needed. Such a system would enable farmers to make informed management decisions, minimize crop losses, and enhance overall agricultural productivity.

Business Objectives

Main Objective

To develop an image classification model that can automatically detect and classify leaf diseases in corn, tomato, and potato plants.

Specific objectives

- To collect and preprocess images of corn, tomato, and potato leaves (both healthy and diseased).
- To train and evaluate a machine learning model capable of identifying disease types based on leaf images.
- To compare model performance using different classification algorithms.
- To provide a simple, user-friendly interface that can assist farmers in identifying crop diseases through image uploads.

Success criteria

- The model achieves an accuracy of 90% or higher on test images.
- It correctly identifies and classifies diseases in all three crops.
- It has a low number of false results (wrong predictions).
- A simple system or interface is developed for easy use by farmers.

DATA & EDA

Data Source

The dataset used in this project is obtained from [Mendeley Data — “PlantVillage Dataset: Plant Leaf Diseases”](#). It contains labeled images of more than one crop leaf, but for this project, we used Potato, corn, and tomato crop leaves, which were categorized into **10 classes**, including both healthy and diseased conditions such as early blight, late blight, rust, and leaf spot. The images are well-structured in folders by class, making them suitable for loading and preprocessing using TensorFlow. The dataset provides a reliable foundation for training a deep learning model to detect and classify plant leaf diseases accurately.

Data Description

The dataset used in this project contains a total of 15,848 labeled images of corn, potato, and tomato leaves, collected under both field and controlled lighting conditions. All images are stored in JPEG format and resized to 224×224 pixels for model training to ensure consistency.

The dataset is organized into 10 distinct classes, representing both healthy and diseased leaf conditions:

- *Corn___Cercospora_leaf_spot (Gray_leaf_spot)*
- *Corn___Common_rust*
- *Corn___Northern_Leaf_Blight*
- *Corn___healthy*
- *Potato___Early_blight*
- *Potato___Late_blight*
- *Potato___healthy_aug*
- *Tomato___Early_blight*
- *Tomato___Late_blight*
- *Tomato___healthy*

This diverse dataset provides a strong foundation for training and evaluating a deep learning model capable of identifying and classifying multiple crop leaf diseases accurately.

Data Abnormalities

The data had 14 duplicated images that were removed before preparing the data further for modelling. It also had an imbalance in the classes, where some had less data(images) and others had more data(images).

Exploratory Data Analysis

Random images found in the dataset.

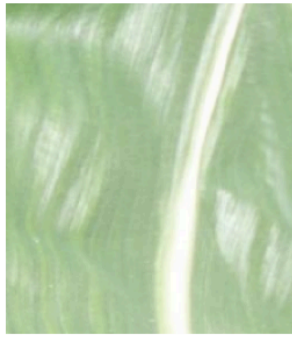
Potato__Early_blight



Corn__Northern_Leaf_Blight



Corn__healthy



Tomato__Late_blight



Corn__Northern_Leaf_Blight



Corn__Northern_Leaf_Blight



Tomato__healthy



Potato__Late_blight



Corn__Cercospora_leaf_spot Gray_leaf_spot



Corn__Northern_Leaf_Blight



Tomato__Late_blight



Potato__Late_blight



Corn__Common_rust



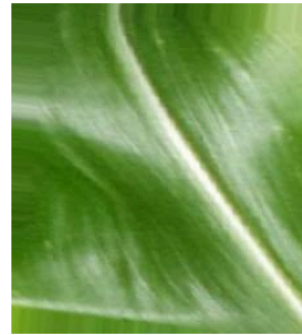
Potato__Late_blight



Corn__healthy



Corn__healthy



Distribution of class counts

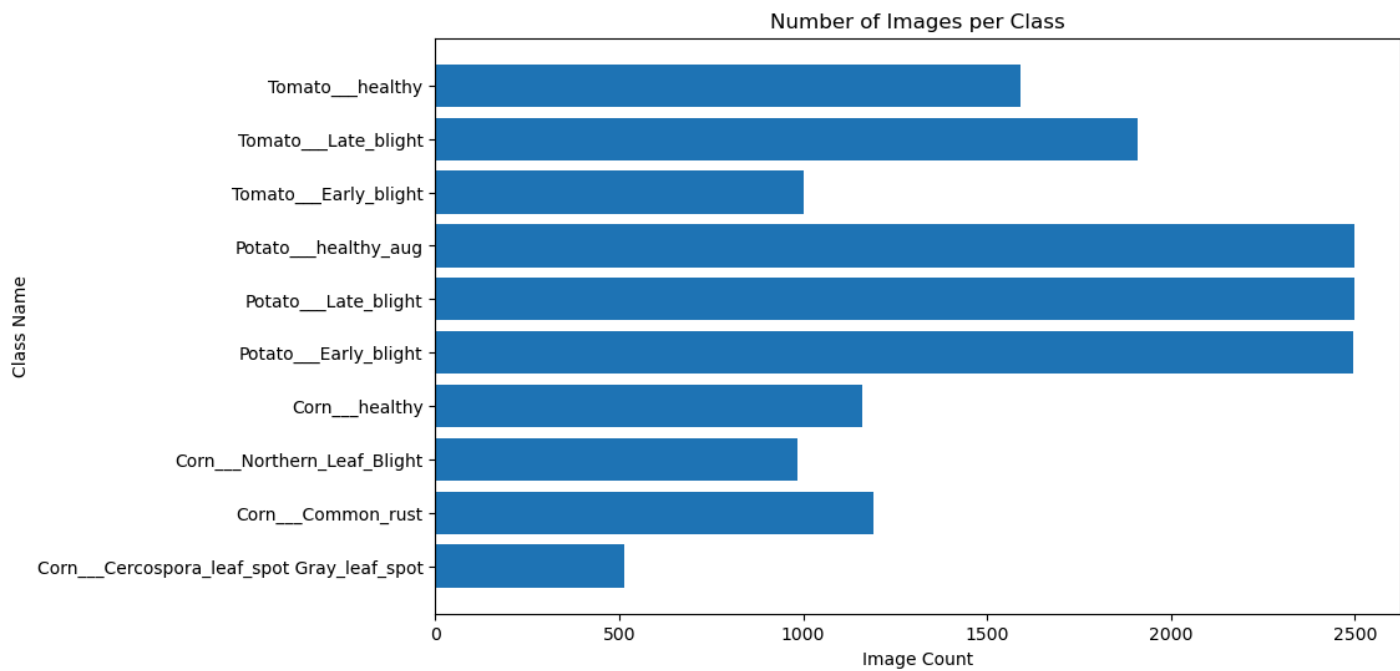
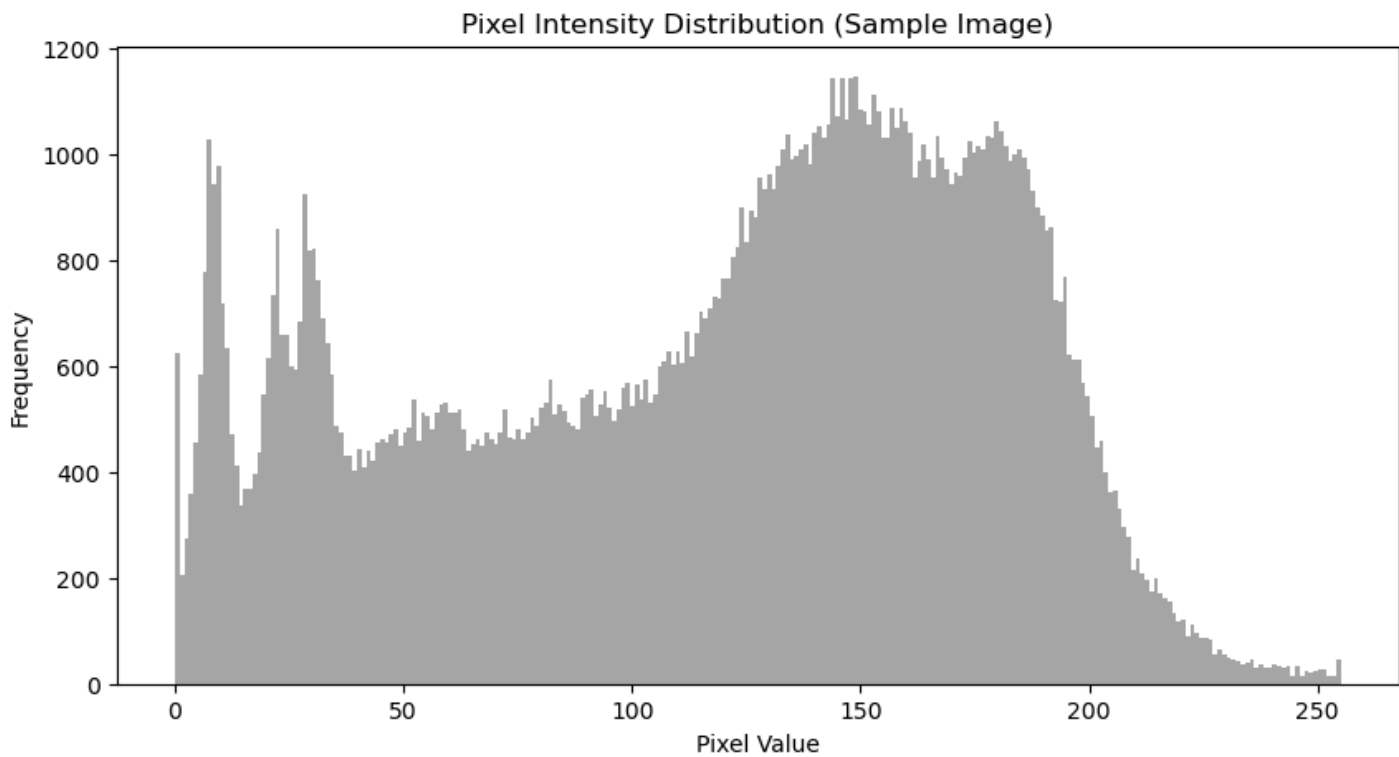


Image pixel intensity Distribution



DATA PREPARATION

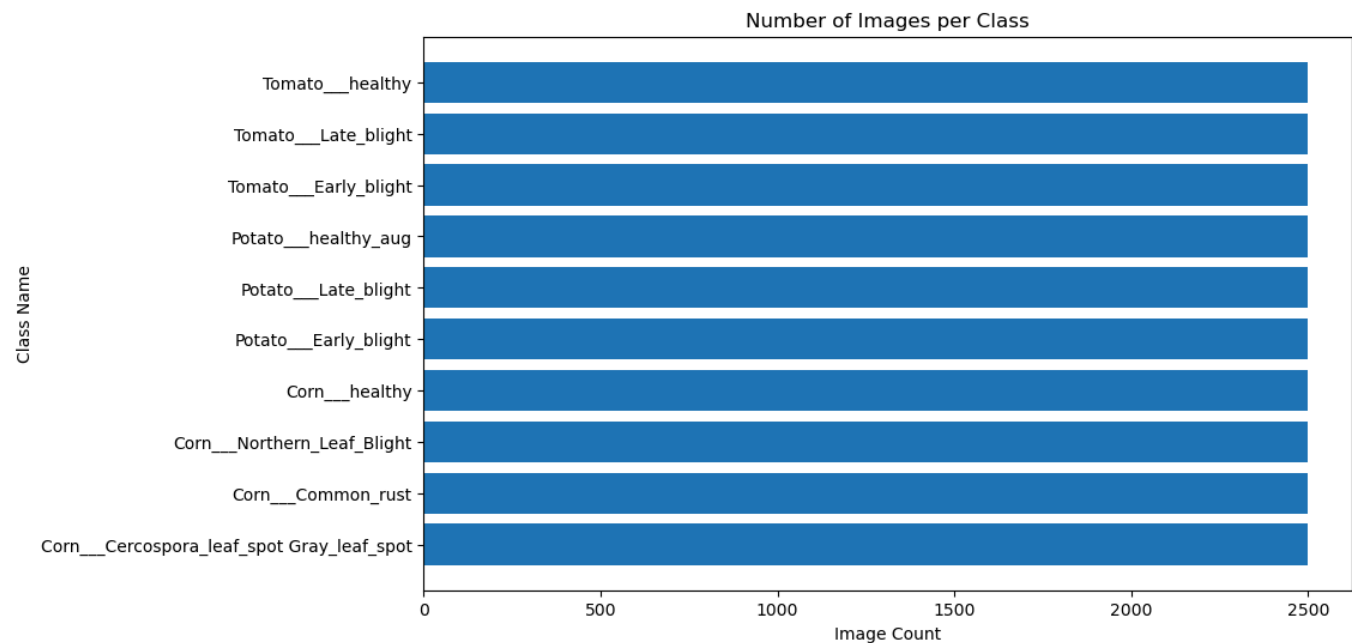
Data Cleaning

To ensure dataset quality, 14 duplicate images were identified and removed to prevent redundancy and model bias. A significant class imbalance was then addressed in the data preparation part before modelling.

Data preparation

The dataset containing 25,000 images across 10 classes was split using TensorFlow's `image_dataset_from_directory()` function. A 70–30 split was applied, allocating 17,500 images for training and 7,500 for validation and testing. Images were resized to 224×224 pixels, and batched in groups of 32 for efficient processing. A fixed random seed (123) ensured reproducibility. The validation set was further divided equally into validation and test sets, providing a balanced and reliable structure for model training and evaluation. To enhance model generalization and prevent overfitting, data augmentation was applied using TensorFlow's **Sequential API**. The process introduced random flips, rotations, and zooms ($\pm 10\%$), enabling the model to learn from varied image orientations and lighting conditions for better real-world performance.

Distribution of classes after augmentation



MODELLING

CNN (custom-built Convolutional Neural Network)

Model Architecture

The model architecture was a custom CNN built using the Keras Sequential API. The network begins with a rescaling layer (1./255) for normalization, accepting an input shape of (224, 224, 3). The core consists of a convolutional base with three Conv2D layers, with filter sizes increasing from 32 to 64, and then to 128. Each convolutional layer is followed by a MaxPooling2D layer to downsample feature maps. The classifier head flattens the output, passes it through a Dense layer of 128 units, applies a Dropout rate of 0.5 for regularization, and concludes with a 10-unit Dense output layer using 'softmax' activation.

Training Process and Tuning

The model was compiled using the 'adam' optimizer for its adaptive learning rate and 'sparse_categorical_crossentropy' as the loss function, with 'accuracy' as the primary metric. Key hyperparameters were selected to prevent overfitting, including a Dropout rate of 0.5 and a Dense layer size of 128 units, which provided sufficient capacity without being computationally excessive. To combat the identified class imbalance, a dictionary of class weights was supplied during training to apply a higher penalty to misclassifications of minority classes.

MobileNetv2 (Transfer learning model)

Model Architecture

This model utilized transfer learning, employing the **MobileNetV2** architecture pre-trained on the ImageNet dataset. The base model was loaded with its top classification layer removed (`include_top=False`), and its layers were **frozen** (`base_model.trainable = False`) to act as a fixed feature extractor. A custom classifier head was added, starting with a `GlobalAveragePooling2D` layer, followed by a Dense layer of 128 units, a Dropout layer with a 0.3 rate, and a final 10-unit Dense layer with 'softmax' activation.

Training Process and Tuning

The key hyperparameter decision was the transfer learning strategy: **freezing the base model** to leverage its pre-learned features. The custom head was tuned with a Dense layer of **128 units** and a Dropout rate of **0.3**. The model was compiled with the 'adam' optimizer and 'sparse_categorical_crossentropy'. Training was set for a maximum of 15 epochs but was dynamically regulated by an **EarlyStopping callback**. This callback monitored 'val_loss' with a patience of 3, automatically stopping the training at the best-performing epoch to prevent overfitting.

MODEL EVALUATION

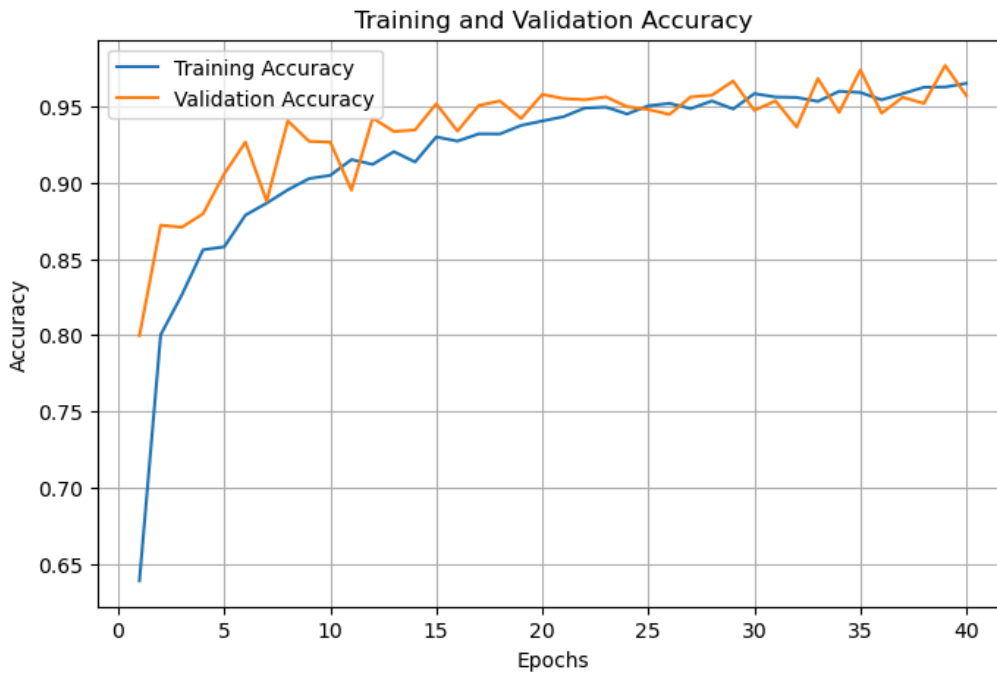
CNN

Results and findings

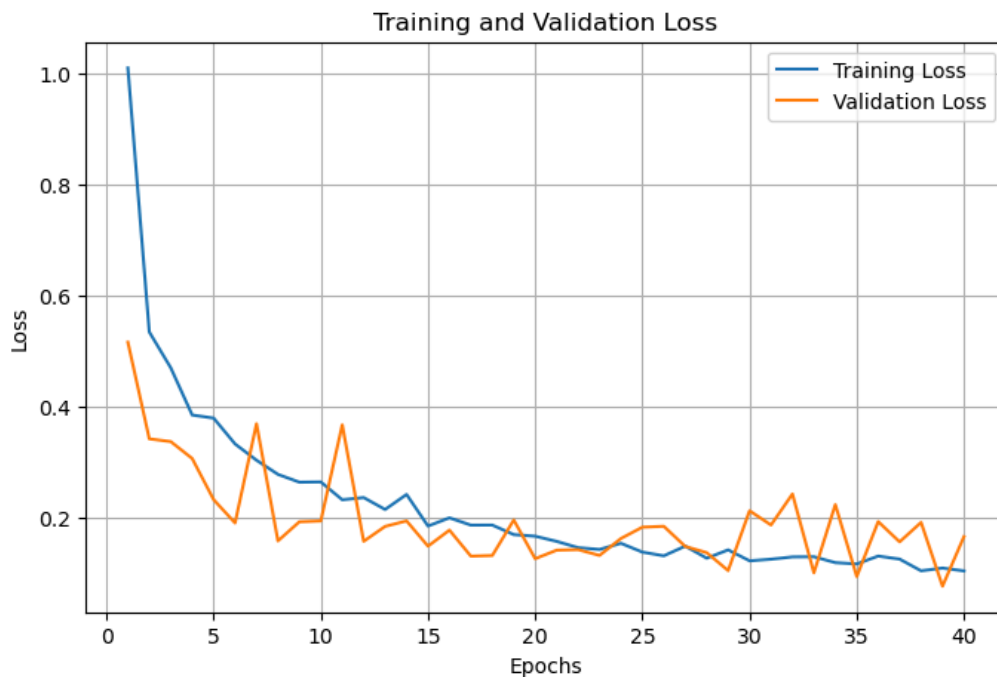
The CNN model demonstrated strong learning performance over 40 epochs. Training accuracy steadily increased from around **65%** to approximately **95%**, while validation accuracy closely followed, stabilizing between **93–96%**. The close alignment between the two curves indicates that the model generalized well to unseen data without significant overfitting. Similarly, the training and validation loss curves showed a clear downward trend. Training loss decreased sharply from about **1.0** to below **0.15**, while validation loss stabilized between **0.15 and 0.20**. This consistent reduction in loss confirms that the model effectively minimized prediction errors and achieved stable convergence.

This strong generalization is confirmed by the model's performance on the final test set, where it achieved an outstanding overall **accuracy of 96%**. The **confusion matrix** provides a detailed breakdown of this high-level performance. Where the model excelled at identifying healthy plants, however, the confusion matrix also quantifies the model's primary challenge: distinguishing between visually similar diseases. The most significant area of confusion was between corn diseases, where 71 images of `Corn_Northern_Leaf_Blight` were misclassified as `Corn_Cercospora_leaf_spot`. A similar, though less pronounced, confusion occurred between tomato diseases, with 25 images of `Tomato_Late_light` being mistaken for `Tomato_Early_blight`. Despite these specific inter-class difficulties, overall, the CNN model exhibited high accuracy, low loss, and strong generalization, making it reliable for classifying and detecting plant leaf diseases across multiple crop types.

Training accuracy vs validation accuracy



Training loss vs Validation loss



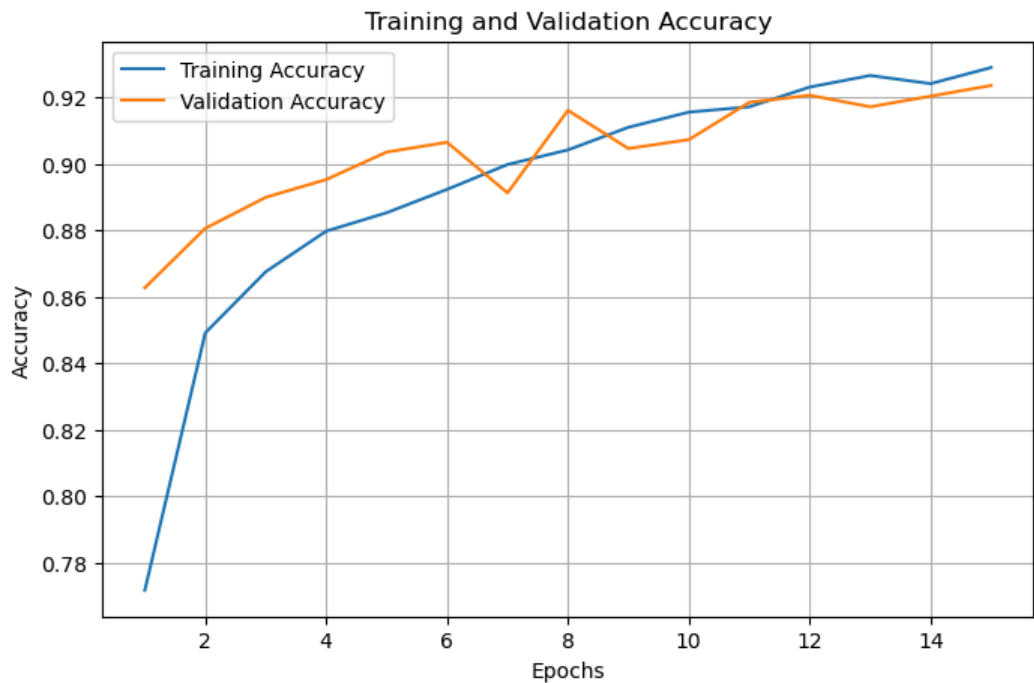
MobileNetv2 (Transfer learning model)

Results and Findings

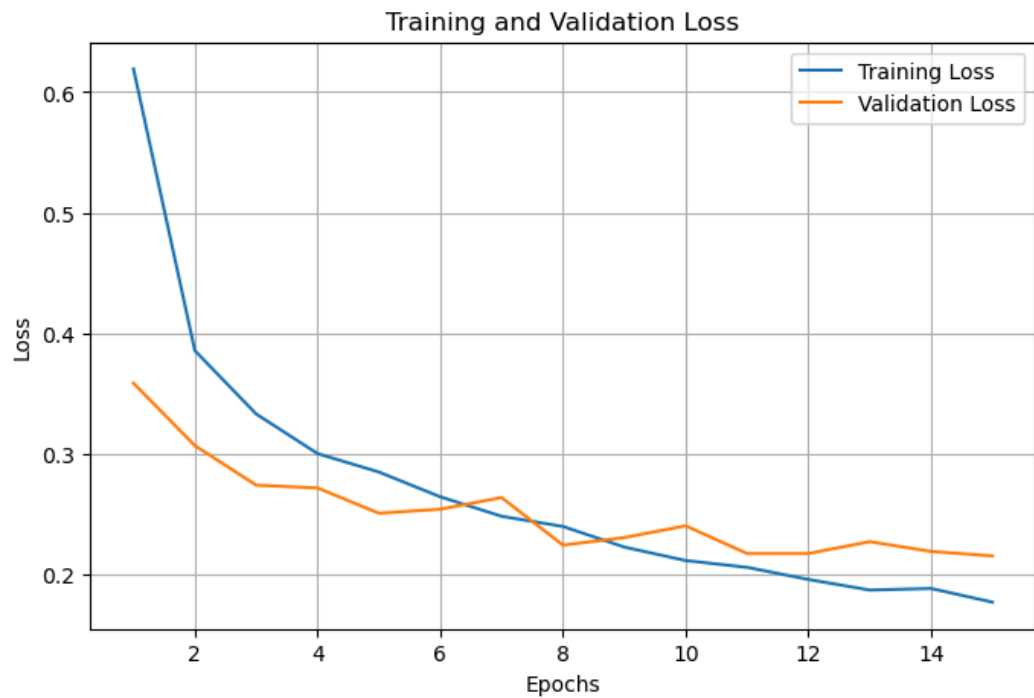
The model demonstrated strong performance and successful training. This is confirmed by the classification report, which shows a robust overall accuracy of 92% on the test set. The model's strength is further reflected in its balanced macro and weighted averages for precision, recall, and f1-score, all registering at 0.92. This indicates that the balancing techniques (class weights and data augmentation) were effective. The training and validation plots support this, showing that accuracy converged above 92% while loss descended to a low, stable value. The close tracking of

the training and validation curves confirms the model is robust, generalized well, and avoided overfitting. A deeper look at the report shows near-perfect identification of healthy classes (e.g., Potato__healthy_aug at 1.00 f1-score), though it exhibited some difficulty distinguishing between more visually similar diseases, such as Corn__Northern_Leaf_Blight (0.81 f1-score) and Corn__Cercospora_leaf_spot (0.83 f1-score).

Train accuracy and validation loss



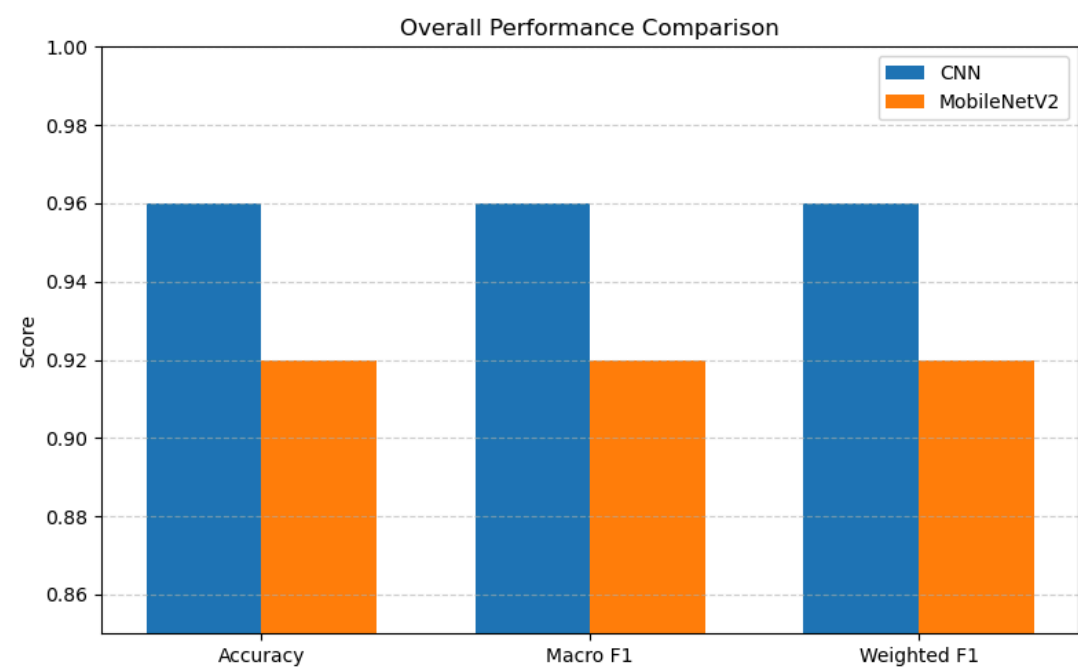
Training loss and validation loss



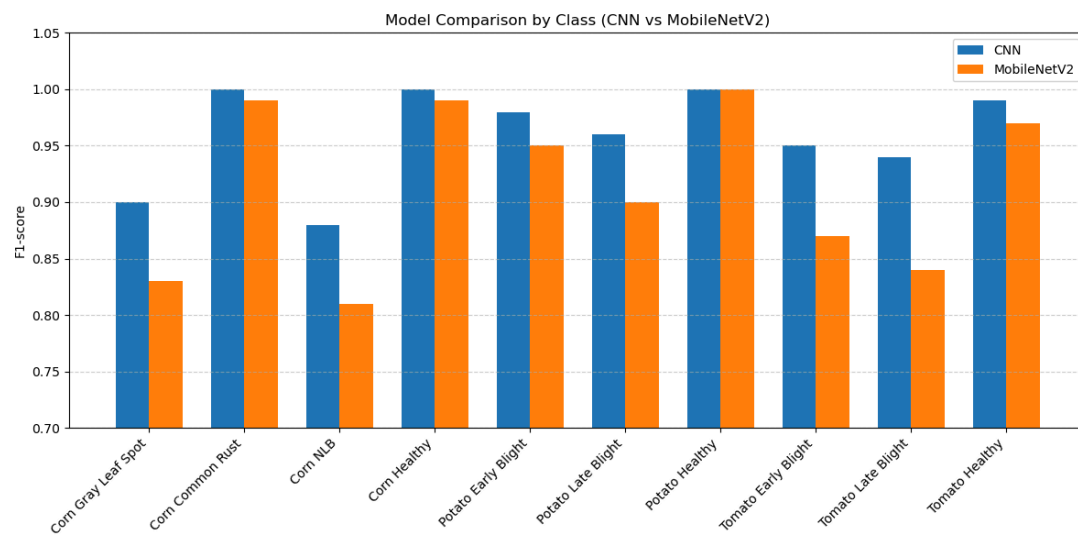
Model comparison

The custom CNN was the superior model, achieving a consistent 96% in accuracy and F1-scores, outperforming the MobileNetV2's 92% on all overall metrics. The per-class F1-score comparison reveals *why*: while both models excelled at identifying healthy leaves, the custom CNN was significantly more accurate at distinguishing between visually similar *diseased* classes, such as Corn NLB and Tomato Late Blight. This indicates the custom architecture learned more robust features for this specific dataset.

Overall performance



Model comparison by class



DEPLOYMENT

The **custom CNN** was selected for deployment due to its superior **96% accuracy** and F1-score. However, a key limitation must be noted: the model was trained *only* on the project's dataset. Its performance on "in-the-wild" images from different cameras or lighting conditions is not guaranteed. As noted in **Future Work**, the next critical step is to retrain the model with a more diverse dataset, including real-world field images, to ensure its practical reliability.



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CONCLUSION

In this project, two models, a custom Convolutional Neural Network (CNN) and a MobileNetV2 transfer learning model, were trained and evaluated on a plant disease dataset containing 10 distinct crop disease classes.

Each class had a balanced number of images (2,500), ensuring fair representation across categories.

Model	Accuracy	Macro F1	Key Strengths
CNN	0.96	0.96	Strong generalization, high recall, and precision across all classes
MobileNetV2	0.92	0.92	Lightweight and efficient, but slightly lower recall for some disease classes

The CNN outperformed MobileNetV2 in both accuracy and F1-score, indicating that the custom architecture was able to learn more robust features specific to this dataset.

All models achieved near-perfect performance on healthy leaf classes, with most errors occurring in visually similar disease types (e.g., Corn Northern Leaf Blight vs Cercospora Leaf Spot).

Recommendation

Model Choice

The CNN model should be selected for deployment since it achieves the highest accuracy (96%) and balanced precision-recall performance.

Practical Application

Deploy the CNN model via Streamlit for farmers or agronomists to upload leaf images and receive:

- The predicted disease class.
- Recommendations on treatment or management.

Future Work

- Extend the dataset to include more plant species and real-world images that include varied lighting and backgrounds.
- Implement Grad-CAM visualizations to highlight which leaf areas the model focuses on.
- Consider edge deployment (e.g., TensorFlow Lite) for use in mobile devices.
- Train the Model with external images (e.g., Google images and field work images)